

Architecture — PA Audit PWA

Data Flow (offline-first)

```
flowchart LR
    U[User Input] --> SC[PII Scrubber<br/>per-install HMAC, no token map]
    SC --> AE[Forensic Audit Engine<br/>structured RuleIds]
    AE --> DB[(Dexie IndexedDB<br/>TheValleyDB_PA_Audit v2)]
    DB --> OB[Outbox<br/>++seq ordering, stable idempotency key]
    OB --> W[signed POST<br/>HMAC-SHA256 + timestamp| W[CF Worker Dispatcher]]
    W --> IN[KV dedupe<br/>ack: 24h TTL| IN[Ingestion Endpoint]]
    IN --> OB
    IN --> W
    W --> IN
    IN --> OB
```

Contract Notes

- Client deletes outbox op only after 2xx (delete-after-ACK)
- 409 = deferred, honors Retry-After, retryCount not incremented
- KV replay suppression ≠ exactly-once; ingestion receives Idempotency-Key
- Privacy gate: eslint `no-restricted-syntax` blocks `originalValue` + static salts, required CI check

Scope Note

This replaces the enterprise-scale reference diagrams (service mesh, ML pipelines, PQC rollout gantt) with the one flow that actually exists in this repo: a single-operator PWA writing to IndexedDB, syncing through a signed outbox to a Cloudflare Worker. Nine aspirational diagrams would misrepresent the system; this one matches it.